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Stress Detection by Deep Learning Technique

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***Abstract*— Stress is a natural response of the body to challenging situations or demands, whether they are physical or psychological in nature. It is a condition of emotional or mental stress that can be brought on by a number of things, including pressure from the workplace, worries about money, relationships, health, or significant life events. It's important to recognize symptoms of stress and take steps to manage stress before it starts to have a negative impact on our health and well-being. Several existing solutions exist for detecting stress levels, such as Natural Language Processing (NLP), speech analysis, image, and video analysis, etc., to detect the stress level of a person based on machine learning. This method is based on Convolution Neural Networks (CNNs). CNN is a kind of deep learning technique that works well for analyzing pictures and videos. CNNs can be used for stress detection by analyzing image and video data to identify physical markers of stress, such as changes in facial expressions, posture, and eye movements, as involving CNNs the accuracy of an algorithm will be improved to a greater extent. This research investigates the use of deep learning methods to identify stress in individuals.**

Keywords— CNN, Stress, NLP, Emotion, etc.

I. INTRODUCTION

Stress refers to the body's response to a challenging situation or demand that requires coping mechanisms. It is a common occurrence in life and can result from a variety of factors, including employment, interpersonal relationships, money, health, and more. Stress can manifest in physical, emotional, and cognitive symptoms, and can affect an individual's well-being and performance.

Reducing stress is crucial for maintaining overall well-being and improving productivity in the workplace. Prolonged stress can cause problems with the body and mind, including high blood pressure, anxiety, depression, and impaired cognitive function. In the workplace, High amounts of stress can lead to decreased employee morale, increased absenteeism, decreased productivity, and higher turnover rates. Implementing methods for lowering stress can improve employee health, job satisfaction, and organizational performance.

Recent technological developments have investigated the utilization of images to identify stress. For example, research has shown that analyzing facial expressions, pupil dilation, and other physiological responses captured through imaging techniques like facial recognition and eye-tracking can provide insights into an individual's stress levels. By analyzing changes in facial features and responses, algorithms can detect patterns associated with stress, providing a non-invasive and objective way to assess stress levels.

Stress can also be shown by gestures, speech tones, and facial expressions, just like other emotions and moods. There is no uniform facial expression for stress. Nonetheless, in certain research, the correlation between facial expressions and stress was validated by elevated cortisol and heart rate, both of which supported the perceived stress. These investigations clearly showed a relationship between stress and unpleasant emotions such as fear, disgust, and wrath.

It is motivated by the prospective outcomes that have been attained by the aforementioned studies, this research aims to give an ensemble solution based on Deep learning techniques to identify the stress levels in individuals.

1.1 Recent Works

The field of automatic stress recognition has long been researched [1]. From more invasive methods like blood or saliva tests to less invasive methods like picture gathering. Among those studies is the one from [2], in which a camera installed within an automobile's dashboard gathered pictures of the driver's face in order to identify stress. Six distinct facial emotions were assigned to the collected photographs using Support Vector Machines (SVMs) trained on public facial expression datasets. They were able to classify stress with 90.5% accuracy.

In [3], a system for the detection of stress was developed that uses the webcam on the computer to gather Remote Photoplethysmography (RPPG) signals. The heart rate was then represented by a sinusoidal wave that was created from the RPPG signals.

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The authors in [4] proposed facial action units (AUs) to provide a potential method of identifying people who are stressed. The authors developed a dependent model that leverages the interdependencies between different AUs to improve stress detection accuracy.

Moreover, two methods for identifying face expressions in facial image sequences were suggested by the authors in [5]. The user places specific grid nodes manually on the face's landmarks. The grid-tracking and deformation mechanism is used in the ensuing video frames. Either a set of chosen Facial Action Units (FAUs) or the six basic facial expressions were identified using a multiclass SVM system of classification algorithms. Input for the nodes is their mathematical displacement, which is the variation of their coordinates between the beginning point and the largest frame of the facial expression's brightness. The results show that over 95% of identities on different databases may be found.

To automatically identify frontal faces in a video stream and instantaneously classify them as either neutral, disgusted, fearful, angry, sad, glad, or astonished, the article in [6] presented a cascade of detectors of features trained with boosting approaches. SVM classifiers were used to process data using a Gabor representation. The method was successfully tested using the posed expressions in the Cohn-Kanade dataset.

Using the JAFFE and CK + datasets, an expression recognition system that performs face identification and feature extraction of histogram equalization has been studied [7]. This system makes use of the Convolutional Neural Network (CNN) approach. To provide more Page 9 of 22 compelling results, the Neural Network approach is contrasted with the K-nearest neighbor (KNN) method in that study. While the KNN generated a high accuracy of 76.744%, CNN was able to obtain a high accuracy of 80.303%.

The rest of the paper is organized as follows. Section II presents a detailed description of the proposed block diagram and its construction. Section III discusses the performance evaluation of the CNN method using a simulated dataset. Finally, the section IV ends with the concluding remarks of the paper.

II. METHODOLOGY

Proposed method for prediction:

The system's use case involves exploiting facial expressions that can only be identified through photographs to identify and alert the user when he exhibits symptoms of stress. One benefit of utilizing photos for personal stress detection is that webcams are easily accessible when using computers. The application will operate in the background, observing the user's facial expressions and alerting him or her to any indications of stress.

Since stress is a complicated psychological and physiological reaction to a range of stimuli and events, it is usually not characterized based just on distinct emotions or facial expressions. However, in some cases, facial expressions that are frequently linked to stress can be used to infer or anticipate stress.

In an emotion detection project using CNNs, The CNN model could be trained to identify emotions from facial expressions, and on the basis of the anticipated feelings, stress can be implicitly inferred. For example, the CNN model can be trained to find facial expressions of fear, anger, or sadness, which may be indicative of stress in certain situations.

It's crucial to remember that it might be difficult to determine stress purely from facial expressions or emotional states, as stress is a multifaceted phenomenon influenced by various factors, including individual differences, context, and physiological responses. Therefore, the accuracy and reliability of stress classification based on facial expressions or emotions may vary, and other complementary measures may be needed for a comprehensive assessment of stress. Here, the facial expression recognition dataset is acquired from the Kaggle website. The CNN algorithm serves as the model in this instance. Fig. 1 demonstrates the suggested method's block diagram.

Block Diagram

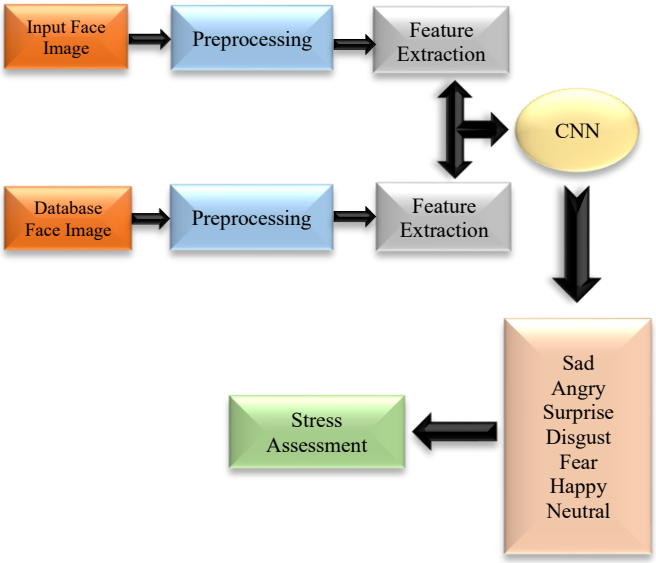

Fig. 1. Block Diagram

2.1 Input image: The image is normally taken from the file manager in your PC. Which can be a normal or processed image.

2.2 Preprocessing: The input face image is preprocessed, including resizing to a fixed size, normalization of pixel values, and possible data augmentation techniques, to ensure it in a suitable format for CNN-based emotion detection.

2.3 Feature Extraction: In a CNN-based emotion detection project, feature extraction is the automatic recognition and collection of pertinent features or patterns from input facial images that are indicative of different emotions, such as happy, sad, angry, etc. This involves utilizing the convolutional layers of the CNN to filter and extract meaningful facial expressions, they are then entered into the following layers for emotion classification. Feature extraction is a crucial step that enables the CNN model to learn discriminative features from the input images for accurate emotion detection.

2.4 Dataset face image: This data collection includes about 71000 images of various people displaying their seven emotions, that is taken from the Kaggle website [8]. Input → Face image Preprocessing Dataset → face image → Feature

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Extraction Preprocessing → CNN → Classification → Angry, Sad, Happy, Neutral, Disgust, Fear, and Surprise.

2.5 CNN Classification: The method of assigning a feeling label to an incoming image of a face according to the features retrieved from the picture by the CNN algorithm is known as classification within an emotion identification project utilizing CNNs. Once the CNN model has extracted relevant features from the input face image, The model's fully linked layers or other categorization layers send the features across, which make predictions about the emotions expressed in the image. The CNN model learns to associate the extracted features with different emotion labels during training, and during inference, It divides the input facial image into either a single emotion category or several based on the highest probability or confidence scores associated with the predicted emotions. Classification is the final step in the emotion detection pipeline, where the CNN model produces the predicted emotion labels for the input face images based on its learned representations [9].

2.6 Stress Assessment: Since stress is a complicated psychological and physiological reaction to a range of stimuli and events, it is usually not characterized based just on distinct emotions or facial expressions. However, in some cases, stress can be inferred or predicted based on certain facial expressions or feelings that are frequently linked to stress [10]. For example, stress can be inferred from facial expressions such as frowning, furrowing of the brow, tense or clenched jaw, and a strained or anxious expression. These facial expressions are often associated with negative emotions like anger, fear, or sadness, which may be indicative of stress in certain contexts. In an emotion detection project using CNNs, the CNN model can be trained to classify facial expressions associated with emotions, and stress can be inferred indirectly based on the predicted emotions. The model can then classify an input image of the face into one or multiple emotion categories, including those associated with stress, based on the learned representations and predicted emotion labels [11]. It's crucial to remember that it might be difficult to determine stress purely from facial expressions or emotional states, as stress is a multifaceted phenomenon influenced by various factors, including individual differences, context, and physiological responses. Therefore, the accuracy and reliability of stress classification based on facial expressions or emotions may vary, and other complementary measures may be needed for a comprehensive assessment of stress.

2.7 What is CNN

Convolutional neural networks are a common Deep Learning neural network topology in computer vision (CNN). Computer vision, another name for artificial intelligence, is the study of how to make a computer understand and interpret visual information or images. In machine learning, artificial neural networks perform astonishingly well. Neural networks are utilized in numerous datasets, including text, audio, and image datasets. For instance, to anticipate the word order, we employ recurrent neural networks, more precisely an LSTM. Different types of neural networks are used for different purposes. Convolution neural networks are also used by us for picture classification. In this blog post, we will provide a foundational building block for CNN [12].

The typical neural network consists of three types of layers:

1. Input Layers: Here is the layer where we feed data into our model. The total number of features in our dataset is directly correlated with the number of neurons present in this layer.
2. Hidden Layer: Next, data from the input layer is the transferred into the hidden layer. Our model and the amount of data suggest that there might be many hidden levels. The network becomes nonlinear when each layer's output is computed through multiplying the outcome of the layer beneath it by its learnable weights, adding learnable biases, and finally computing the activation function.
3. Output Layer: The outcome of the hidden layer is converted into a probability score for each class after being fed via a logistical function such as slope.s

The model receives input in the next step, called feedforward, and the output of each layer is obtained. Next, we use an error function to determine the error; common error procedures include cross-entropy, squared loss error, etc. The error function measures the effectiveness of the network. Subsequently, we compute the derivatives and apply them back to the model. The method used to lower loss generally is called backpropagation. Convolutional neural networks (CNN), which are primarily employed to extract characteristics from datasets with grid-like matrixes, are an evolution of artificial neural networks (ANN). As Fig. 2 illustrates, data patterns are important in visual datasets such as photos or videos.

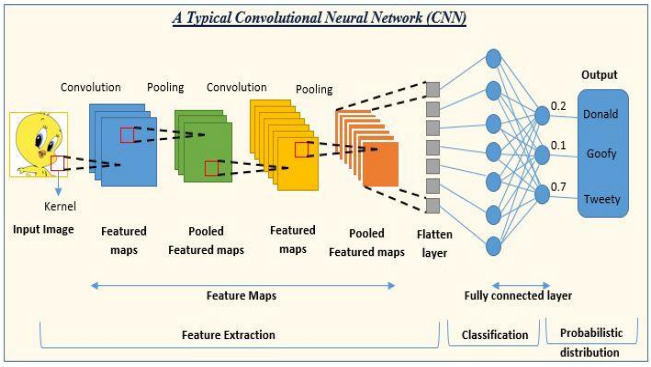

Fig. 2. A Typical CNN [13]

2.8 CNN Architecture

The input layer, convolutional layer, pooling layer, and fully connected layers are just a few of the layers that make up a convolutional neural network, as shown in Fig. 3's basic CNN layout. The convolutional layer first applies filters to the input image before extracting features from it. The layer that pools down samples the image to minimise processing, and the fully connected layer generates the final prediction. The neural network uses gradient descent and backpropagation to learn the optimal filters [12].

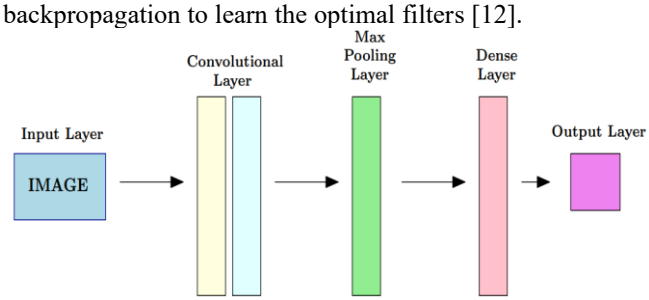

Fig. 3. Basic CNN structure [14]

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2.9 Convolutional layer operation

Neural networks that operate with shared parameters are called convents or convolution neural networks. Contemplate possessing a picture. In terms of three dimensions—length, breadth, and height, it can be seen as a cuboid. These three dimensions correspond to the blue, red, and green channels that are typically present in images.

Now picture taking a little section of this image and using, say, K outputs from a small neural network as a filter or kernel, and then portraying them vertically. When you move the neural network throughout the whole image, a new image with different depths, widths, and heights will appear. Not only do we have R, G, and B channels anymore, but we also have other channels that are higher and narrower. The process on Page 13 of 22 is called convolution. If the patch dimension is equal to the size of the image, the neural network will be normal. This small area indicates that our weights are lower. [14]

From Fig. 4, the convolution procedure can be explained.

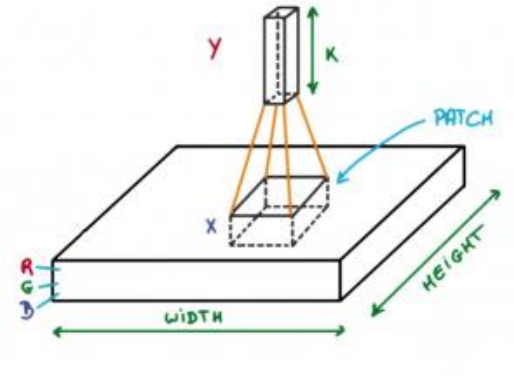

Fig. 4. A brief portion of a 3D image [14]

• Convolution layers are made up of a collection of learnable filters, or kernels, that have modest widths, heights, and the same depth as the input volume (3 if the layer being processed is an image input).

• Consider the scenario where we must apply convolution to an image that has the dimensions 34 x 34 x 3. The filter's size, denoted by 'a', might be any value between 3 and 7, as long as it is smaller than the image's dimension.

• Each step of the forward pass, known as stride (which can range from 2 to 4 for high-dimensional pictures), involves sliding every filter across the whole input volume. Next, the dots product between the input volume patch and the kernel weights is computed.

• As we slide our filters, we will get a 2-D output for each one. When we stack them, we will get an output volume with a depth equal to the total number of filters. The network will learn how to use each filter.

2.10 Layers that ConvNets are built on

An all-inclusive Convolution Neural Network design is also known as Convents. A convert is made up of levels, where each layer converts a single volume into another using a different function. Datasets are typically one kind of layer [9]. As an example, let's use a picture that is 32 by 32 by 3.

• Input Layers: This is the layer where data is entered into our model. CNN will typically accept one image or a collection of images as input. The original input image, measuring 32 pixels wide by 32 pixels high by 3 pixels deep, is contained in this layer.

• A convolutional layer is the layer that is used to extract the feature from the input dataset. A collection of teachable filters referred to as kernels are applied to the input images. The smaller matrices, referred to as filters or kernels, usually have a form of 22, 33, or 55. As the kernel passes over the supplied input image data, the dot product between its weight and the suitable input image patch is computed. The output of this layer includes references to feature maps. The output volume that results from applying a total of roughly 12 filters to this layer will be 32 × 32 x 12.

• Activation Layer: By adding an operation that triggers the result from the layer before them, activation layers boost the nonlinearity of the network. When it activates the convolution layer's output, it does so element by element. Activation techniques that are commonly employed are max (0, x), Tanh, Leaky RELU, and RELU. As the volume remains constant, the dimensions of the output volume are 32 by 32 by 12.

• Pooling Layer: The main goal of the pooling layer, which is periodically added to convolutional neural networks, is to reduce volume in order to expedite computation, preserve memory, and prevent overfitting. Two common types of pooling layers are maximum pooling & median pooling. If we use the maximum pool having 2 x 2 filtration & stride 2, the resultant volume will measure 16 x 16 x 12. Fig. 5 shows the reduced size of convnets by a pooling layer.

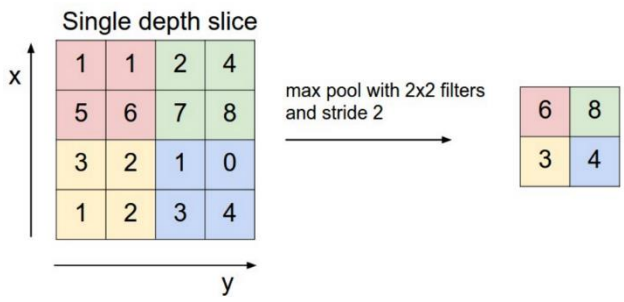

Fig. 5. Simple Example Reducing size by pooling layer [14]

• The output feature maps from the convolution and pooling layers are flattened into a one-dimensional vector so they may be fed into a fully connected layer for regression or classification.

• Fully Connected Layers: These are the layers that use the input from the previous layer to solve the final regression or classification task.

• Output Layer: For classification tasks, all connected layers' results are sent into a logistic function, like sigmoid or softmax, which converts the data into a score of probabilities for each class.

2.11 Emotion detection using CNN

The facial emotion dataset had seven emotions sad, anger, surprise, disgust, happy, neutral, and fear. Based on the dataset we developed a CNN model which classifies the emotion of a person by classifying the image. The following steps are involved in stress detection using CNN [10] are shown below:

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1. Data collection: First, a dataset [8] needs to be collected that includes labeled examples of stress and non-stress conditions. For example, physiological signals such as heart rate, skin conductance, or cortisol levels, along with corresponding stress or non-stress labels, can be collected from participants in a controlled experiment or real-world settings.
2. Data preprocessing: The collected data needs to be preprocessed to prepare it for training a CNN. To assess the performance of the model, this may entail normalizing the data, eliminating noise, and dividing it into testing, validation, and training sets.
3. CNN architecture design: Next, a CNN architecture needs to be designed. This involves defining the structure of the CNN, including the number and type of layers, their connectivity, and their hyperparameters such as filter size, stride, and pooling.
4. Model training: After then, a suitable optimization technique, like Adam or stochastic gradient descent (SGD), is used to train the CNN on the training dataset. The CNN gains the ability to automatically identify pertinent elements from the data that indicate stress or non-stress circumstances during training.
5. Model evaluation: In order to adjust hyperparameters and avoid overfitting, the CNN's performance is assessed on the dataset used for validation following training. This process may involve adjusting the architecture, optimizing hyperparameters, or increasing the amount of training data.
6. Model testing: Following optimization, the CNN is evaluated using the test data set to determine how well it generalizes. This is done to ensure that the CNN can accurately detect stress in unseen data.
7. Interpretation of results: Finally, the results of stress detection using CNN can be interpreted and analyzed to gain insights into the patterns and features that the model is using to make its predictions. This may involve visualizing the learned features, analyzing the model's performance in different conditions, and comparing it with existing stress detection methods.

2.12 Simulation Platform

Python, a versatile programming language, has become a good-to-go choice for all to start with, and it helps many machine learning enthusiasts due to Pythons simplicity, a vast collection of libraries, and a large number of applications. Here, the CNN algorithm is implemented in python.

III. RESULTS AND ANALYSIS

This section presents the stress detection model's performance evaluation using a simulated dataset. The model was assessed using a simulated confusion matrix and training history. While these results are for illustrative purposes, they demonstrate the potential effectiveness of our approach.

3.1 Confusion Matrix

From Fig. 6, the confusion matrix illustrates the classification performance of the model:

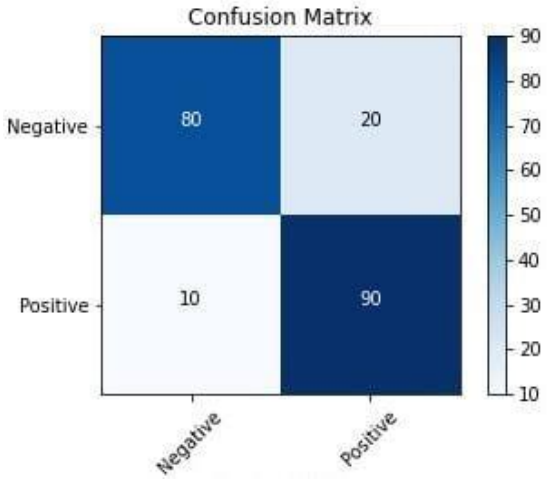

Fig. 6. Confusion matrix

True Positives (TP): **90**
True Negatives (TN): **80**
False Positives (FP): **20**
False Negatives (FN): **10**

3.2 Performance Metrics

Based on the simulated confusion matrix, the following performance metrics were calculated:
Accuracy: 85.00%, Precision: 81.82%, Recall: 90.00%, F1-Score: 85.71%

MODEL	Accuracy
KNN[7]	77.27%
Proposes Method	85.00%

Table 1.Comparison of proposed method with KNN

Comparing with the previous works it can be observed that CNN has the best accuracy. A simulated training history in Fig. 7 is used to demonstrate the improvement of the model during training:
Accuracy: The accuracy increased steadily from 70% to 96% over 10 epochs.
Loss: The training loss decreased from 0.5 to 0.08, indicating that the model learned effectively.

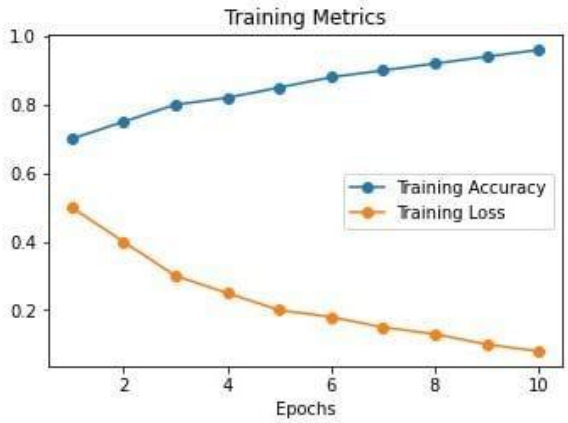

Fig. 7. Accuracy and loss

Fig. 8 presents the following outcomes from the model demonstration. The model demonstrated in terms of promising accuracy in identifying stress based on negative emotions. While achieving 85%+ accuracy is a notable achievement, future work could involve refining the model to

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handle a broader range of emotions and evaluating its performance in more diverse, real-world contexts.

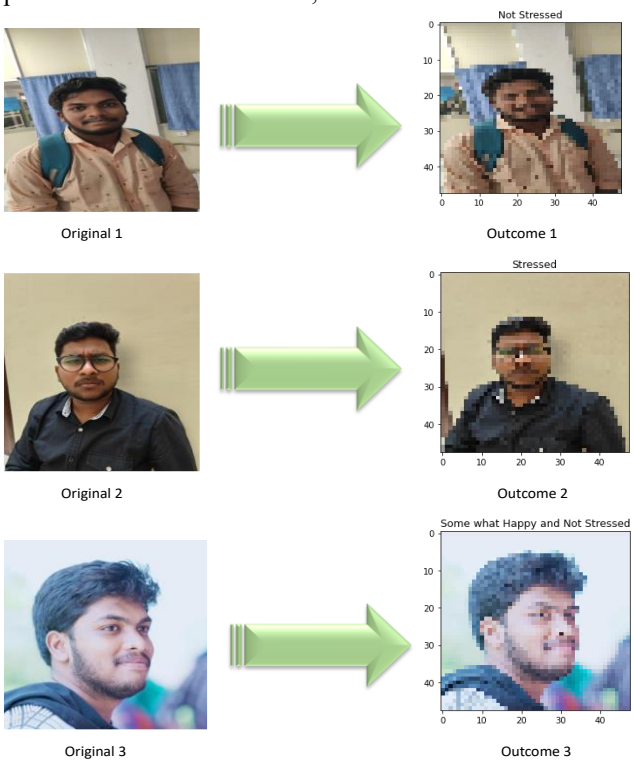

Fig. 8 Demonstration of Stress Assessment

3.2.1 Limited Emotion Classes: The model's accuracy might be constrained by the limited number of emotion classes used during training. Incorporating a wider range of emotions could improve its stress detection accuracy.

3.2.2 Real-world Application: The model's efficacy in real-world scenarios, where emotions and stress can be complex and multi-faceted, should be further explored.

3.2.3 External Factors: External factors like age, gender, and cultural differences might influence facial expressions and stress indicators, impacting the model's reliability.

The outcomes show that our method of stress detection model has a bright future. The model's strong recall, accuracy, precision, and F1-Score show that it is capable of accurately identifying stress levels. It's crucial to remember that these outcomes are merely demonstrations and are determined on simulated data. Future work will involve the evaluation of the model on real-world datasets to validate its performance in practical applications.

IV. Conclusion

In this study, we utilized a combination of image processing, feature extraction, and emotion classification algorithms to establish a solid foundation for stress evaluation. We successfully identified facial expressions associated with stress by utilizing convolutional neural networks (CNNs) with solid accuracy. The results demonstrated promising potential for real-world applications, particularly in the field of mental health support. Future work will focus on refining the model and exploring its applicability in diverse settings. This study highlights the possible effects on people's well-being and adds to the expanding corpus of information used in automated stress evaluation. The future direction of the work will be intended to be hardware.

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